

Sayan Biswas

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Research interests

Decentralized ML focusing on Privacy, Fairness, and Personalization; Trustworthy Distributed Systems; Differential Privacy.

Positions

2023 - present **Postdoctoral Researcher** at EPFL (Scalable Computing Systems Lab) – Lausanne, Switzerland.
Supervisor: Prof. Anne-Marie Kermarrec.

Education

2020 - 2023 **PhD in Computer Science** from INRIA, École Polytechnique, and Institut Polytechnique de Paris – Palaiseau, France.
Supervisor: Prof. Catuscia Palamidessi.

2016 - 2020 **Master of Mathematics (M.Math.)** with **First-Class Honours** from University of Bath – Bath, England.

Grants and Awards

2025 **Young Scholars' Development Program Fellowship** from ACM CCS'25.

2025 **Grant of CHF 65,452.00 from the Federal Office for Defence Procurement of Switzerland and armasuisse** for the project "Privacy-preserving and distributed processing of public data in hybrid trust networks".

2024 **Award for the Best PhD Thesis in Computer Science** from Institut Polytechnique de Paris, France.

Publications

Peer-Reviewed Conferences

- 2026 Sayan Biswas, Anne-Marie Kermarrec, Rafael Pires, Rishi Sharma, Milos Vujanovic: **Communication-Efficient Secure Aggregation in Decentralized Learning**. International Symposium on Reliable Distributed Systems (SRDS) 2026.
- 2025 Kangsoo Jung, Sayan Biswas, Catuscia Palamidessi: **Mitigating Membership Inference Vulnerability in Iterative Federated Clustering Algorithm**. Workshop on Recent Advances in Resilient and Trustworthy Machine Learning-Driven Systems (ARTMAN) co-located with ACM Conference on Computer and Communications Security (CCS) 2025.
- 2025 Jade Garcia Bourrée, Augustin Godinot, Sayan Biswas, Anne-Marie Kermarrec, Erwan Le Merrer, Gilles Tredan, Martijn de Vos, Milos Vujanovic: **Robust ML Auditing using Prior Knowledge**. International Conference on Machine Learning (ICML) 2025. **Accepted as spotlight (top 2.6% of the papers)**.
- 2025 Sayan Biswas, Davide Frey, Romaric Gaudel, Anne-Marie Kermarrec, Dimitri Lerévérend, Rafael Pires, Rishi Sharma, François Taïani: **Low-Cost Privacy-Aware Decentralized Learning**. Privacy Enhancing Technologies Symposium (PoPETs) 2025.
- 2025 Sayan Biswas, Anne-Marie Kermarrec, Alexis Marouani, Rafael Pires, Rishi Sharma, Martijn de Vos: **Boosting Asynchronous Decentralized Learning with Model Fragmentation**. The ACM Web Conference (WWW) 2025. **Selected for oral presentation (top 7% of the papers)**.
- 2024 Sayan Biswas, Anne-Marie Kermarrec, Rishi Sharma, Thibaud Trinca, Martijn de Vos: **"Fair Decentralized Learning"**. IEEE Conference on Secure and Trustworthy Machine Learning (SaTML) 2025.
- 2024 Sayan Biswas, Mathieu Even, Anne-Marie Kermarrec, Laurent Massoulié, Rafael Pires, Rishi Sharma, Martijn de Vos: **Noiseless Privacy-Preserving Decentralized Learning**. Privacy Enhancing Technologies Symposium (PoPETs) 2025.
- 2023 Sayan Biswas, Kangsoo Jung, Catuscia Palamidessi: **Tight Differential Privacy Guarantees for the Shuffle Model with k -Randomized Response**. International Symposium on Foundations and Practice of Security (FPS) 2023.
- 2023 Sayan Biswas, Catuscia Palamidessi: **PRIVIC: A privacy-preserving method for incremental collection of location data**. Privacy Enhancing Technologies Symposium (PoPETs) 2024.
- 2023 Filippo Galli, Sayan Biswas, Kangsoo Jung, Tommaso Cucinotta, Catuscia Palamidessi: **Group privacy for Personalized Federated Learning**. Proceedings of the International Conference on Information Systems Security and Privacy (ICISSP) 2023.
- 2022 Sayan Biswas, Graham Cormode, Carsten Maple: **Impact of Sampling on Locally Differentially Private Data Collection**. Competitive Advantage in the Digital Economy – Resilience, Sustainability, Responsibility, and Identity (CADE) 2022. **Winner of the Best Paper Award**.
- 2022 Sayan Biswas, Kangsoo Jung, Catuscia Palamidessi: **Tight Differential Privacy Blanket for Shuffle Model**. Competitive Advantage in the Digital Economy – Resilience, Sustainability, Responsibility, and Identity (CADE) 2022.
- 2021 Sayan Biswas, Kangsoo Jung, Catuscia Palamidessi: **An Incentive Mechanism for Trading Personal Data in Data Markets**. International Colloquium on Theoretical Aspects of Computing (ICTAC) 2021.

Journals

- 2024 Ugur Ilker Atmaca, Sayan Biswas, Carsten Maple, Catuscia Palamidessi: **A Privacy-Preserving Querying Mechanism with High Utility for Electric Vehicles**. IEEE Open Journal of Vehicular Technology, Volume 5 (2024).
- 2023 Filippo Galli, Kangsoo Jung, Sayan Biswas, Catuscia Palamidessi, Tommaso Cucinotta: **Advancing Personalized Federated Learning: Group Privacy, Fairness, and Beyond**. Springer Nature Computer Science, Volume 4, Issue 6, Article 831 (2023).

Book Sections

- 2021 Kangsoo Jung, Sayan Biswas, Catuscia Palamidessi: **Establishing the Price of Privacy in Federated Data Trading**. Protocols, Strands, and Logic, pp 232-250, LNCS 13066, Springer.

Non-Archival Workshops

- 2025 Jade Garcia Bourrée, Augustin Godinot, Sayan Biswas, Anne-Marie Kermarrec, Erwan Le Merrer, Gilles Tredan, Martijn de Vos, Milos Vujanovic: **Robust ML Auditing using Prior Knowledge**. Technical AI Governance Workshop in conjunction with ICML (TAIG ICML) 2025.
- 2024 Sayan Biswas, Mark Dras, Pedro Faustini, Natasha Fernandes, Annabelle McIver, Catuscia Palamidessi, Parastoo Sadeghi: **Bayes' capacity as a measure for reconstruction attacks in federated learning**. Workshop on Security, Privacy, and Information Theory (Protect-IT) in conjunction with IEEE Computer Security Foundations Symposium (CSF) 2024.
- 2023 Sayan Biswas, Catuscia Palamidessi: **PRIVIC: A privacy-preserving method for incremental collection of location data**. Theory and Practice of Differential Privacy Workshop (TPDP) 2023.
- 2023 Filippo Galli, Sayan Biswas, Kangsoo Jung, Tommaso Cucinotta, Catuscia Palamidessi: **On the adaptive sensitivity of differentially private machine learning**. Privacy-Preserving Artificial Intelligence (PPAI) in conjunction with AAAI 2023.
- 2022 Filippo Galli, Sayan Biswas, Kangsoo Jung, Tommaso Cucinotta, Catuscia Palamidessi: **Group privacy for personalized federated learning**. International Workshop on Federated Learning: Recent Advances and New Challenges (FL-NeurIPS) in conjunction with NeurIPS 2022. **Accepted for oral presentation (top 10% of the papers)**.

Teaching

- Aug'22 - Sep'22 **INRIA-DFKI European Summer School on AI (IDESSAI 2022)**, Saarbrücken, Germany, *Differential Privacy & Federated Learning*: theory and implementation (voted as the *most liked course* of IDESSAI 2022).
- Feb'21 - Jun'23 **École Polytechnique, Palaiseau, France**, CSE 102: Advanced Programming with Python.
- Oct'18 - May'20 **University of Bath, Bath, England**, XX10190: Programming & Discrete Mathematics with MATLAB (2018-19), MA10209: Algebra 1A (2019-20), and MA10212: Prob. & Stat. 1B (2019-20).
- Jul'17 - Aug'17 **Humen Foreign Language School (HFLS), Humen, China**, Taught Mathematics and English at the HFLS International Summer Camp.

Academic Services

- PC member IEEE ICDCS 2026, AISTATS 2026, ACM CCS 2026, IEEE SaTML 2026, ESORICS Workshop on Secure and Trustworthy Machine Unlearning Systems 2025, IEEE SaTML 2025, AAAI Workshop Privacy Preserving Artificial Intelligence 2023
- Ad hoc peer reviewer ACM Transactions on Privacy and Security, IEEE Journal on Selected Areas in Information Theory, IEEE Transactions on Dependable and Secure Computing

Miscellaneous Research Appointments

- Sep'23 - Oct'23 **Visiting Scholar**, Macquarie University, Sydney, Australia. Hosted by Prof. Annabelle McIver and Dr. Natasha Fernandes.
- Jan'22 - Mar'22 **Visiting Scholar**, The University of Warwick, Coventry, UK. Hosted by Prof. Carsten Maple and Prof. Graham Cormode.
- Jun'20 - Aug'20 **Research Intern**, The University of Warwick, Coventry, UK. Supervised by Prof. Graham Cormode and Prof. Carsten Maple.
- Jun'19 - Sep'19 **Research Intern**, INRIA, Palaiseau, France. Supervised by Prof. Catuscia Palamidessi.
- Jun'18 - Aug'18 **Research Intern**, Institute for Mathematical Innovation & University of Bath, Bath, UK. Supervisor: Prof. Christopher Jennison.

Other Achievements

- 2022 Winner of the **Best Paper Award** at the conference on Competitive Advantage in the Digital Economy (CADE) 2022.
- 2018 **Qualified for the International Collegiate Programming Contest (ICPC) European Finals 2018** (*first solver of Problem 5 in the UK and Ireland qualification round*) representing University of Bath.
- 2016 **Honourable Mention in the final round of Indian National Philosophy Olympiad 2016** (*top 10 from India*).
- 2016 **Qualified the Zonal Informatics Olympiad 2016** (*top 6 in West Bengal, India*).
- 2015 **Honourable Mention at the International Linguistics Olympiad Training Camp 2015** (*the invitational round for Indian team selection for IOL 2015*) (*top 15 in India*).
- 2013 & 2014 **Qualified for the Regional Mathematical Olympiad 2013 & 2014** (*top 28 in West Bengal, India*).

Other Work Experience

- Nov'18 - May'20 Lead student-editor of Dept. of Mathematics newsletter of University of Bath
- Jul'19 Invigilator and examiner for the International Mathematical Olympiad (IMO) 2019 at Bath, UK
- Feb'17 - Nov'19 Project coordinator and head for Mathscon (UK's largest student-led maths conference)
- Jan'17 - Mar'20 Volunteer for United Kingdom Mathematics Trust and Mentor for British Mathematical Olympiad